# AN IDEMPOTENT SPLITS THE SPECTRUM INTO THE TWO BASIC OPEN PIECES

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*Allowed standard prerequisites (editorial).*

- The basic open D(f) as the set of primes not containing f. Used at the start of both displayed computations (lines 4004-4008 and 4020-4024). Rewrites the two basic opens in terms of membership of the idempotent in the prime.
- Values of an idempotent in residue fields. Used at the end of both computations (lines 4009-4017 and 4025-4032). Turns non-membership into being equal to 1 modulo the prime and membership into being 0, which makes the two pieces complementary.

**Lemma 0.1.** *Let $R$ be a ring. Let $e \in R$ be an idempotent. In this case*

$$\mathrm{Spec}(R) = D(e) \amalg D(1-e).$$

*Proof.* Note that an idempotent $e$ of a domain is either 1 or 0. Hence we see that

$$\begin{aligned} D(e) &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid e \notin \mathfrak{p}\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid e \neq 0 \text{ in } \kappa(\mathfrak{p})\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid e = 1 \text{ in } \kappa(\mathfrak{p})\} \end{aligned}$$

*Date*: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Similarly we have

$$\begin{aligned} D(1-e) &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid 1 - e \notin \mathfrak{p}\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid e \neq 1 \text{ in } \kappa(\mathfrak{p})\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(R) \mid e = 0 \text{ in } \kappa(\mathfrak{p})\} \end{aligned}$$

Since the image of $e$ in any residue field is either 1 or 0 we deduce that $D(e)$ and $D(1-e)$ cover all of $\mathrm{Spec}(R)$. $\square$